MA3071 - DLI
Financial Mathematics – Section 6
Mean-variance portfolio theory

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Background

- Developed by Harry Markowitz in the 1950s, the theory seeks to mathematically derive optimal portfolios.
- By making some simplifying assumptions, it is possible to develop a rigorous approach to balancing risk and return.
- The output will consist of a set of potential optimal portfolios for an investor to choose from, with the final portfolio selection depending on their preferences.
- This method for portfolio construction focuses on individual security selection, while asset allocation (i.e., the split between asset classes such as equities and bonds) is treated as a separate decision.

Key assumptions

- Investors only consider expected returns and variance of returns over a single time period, making the measure of risk the variance of returns.
- All expected returns, return variances, and return covariances are known for all assets under consideration.
- Investors are never satiated: they always prefer higher returns for a given level of risk.
- Investors are risk averse: for a given level of return, they will always choose a portfolio with a lower level of risk.
- The market is frictionless:
  - there are no taxes or transaction costs.
  - securities are infinitely divisible and can be traded in any (fractional) amount.
  - short selling is permitted in any amount.

Some definitions

- Suppose there are n assets being considered, denoted as $A_1, A_2, \ldots, A_n$, where:
  - R_i is the rate of return on asset A_i.
  - $\mathbb{E}[R_i]$ is the expected rate of return on asset A_i.
  - $\sigma_i^2 = \mathrm{Var}[R_i]$ is the variance of return on asset A_i.
  - $\sigma_{ij} = \mathrm{Cov}[R_i, R_j]$ is the covariance of return between asset A_i and asset A_j. In particular, when $i = j$, $\sigma_{ii} = \sigma_i^2$.
  - $c_{ij} = \frac{\sigma_{ij}}{\sigma_i \sigma_j}$ is the correlation between asset A_i and asset A_j.
  - x_i represents the weight of asset A_i in the portfolio, subject to the budget constraint $\sum_{i=1}^{n} x_i = 1$.
  - R_p is the portfolio return rate such that $R_p = \sum_{i=1}^{n} x_i R_i$.
- ρ represents the risk-free interest rate in the market.

Opportunity set

- The **opportunity set** consists of all possible portfolios, which includes all permutations of assets from the n assets being considered.
- That is a set of portfolios, such that

$$\left\{ R_p \,\middle|\, R_p = \sum_{i=1}^{n} x_i R_i, \quad \sum_{i=1}^{n} x_i = 1, \quad x_i \in \mathbb{R} \right\}$$

Expected return and variance of portfolio

- The expected return and variance of the portfolio are calculated as follows:
  - $\mathbb{E}_p = \mathbb{E}[R_p] = \sum_{i=1}^{n} x_i \mathbb{E}[R_i]$
  - $\sigma_p^2 = \mathrm{Var}[R_p] = \sum_{i=1}^{n} \sum_{j=1}^{n} x_i x_j \sigma_{ij}$
    $= \sum_{i=1}^{n} x_i^2 \sigma_i^2 + 2 \sum_{i=1}^{n} \sum_{j=i+1}^{n} x_i x_j \sigma_{ij}$

Simplest portfolio

- Let us consider the simplest portfolio consisting of only two assets, A_1 and A_2.
- **Question**: What is the opportunity set?

Simplest portfolio, cont.

- The opportunity set is

$$R_p = x_1 R_1 + x_2 R_2, \quad x_1 + x_2 = 1, \quad x_1, x_2 \in \mathbb{R}$$

- Let $x_1 = x$ and $x_2 = 1 - x$. The opportunity set can be rewritten as

$$R_p = xR_1 + (1-x)R_2 = (R_1 - R_2)x + R_2$$

- Then in this case, we have

$$\begin{aligned}
\mathbb{E}_p &= x\mathbb{E}[R_1] + (1-x)\mathbb{E}[R_2] \\
\sigma_p^2 &= x^2\sigma_1^2 + (1-x)^2\sigma_2^2 + 2x(1-x)\sigma_{12} \\
&= x^2\sigma_1^2 + (1-x)^2\sigma_2^2 + 2x(1-x)\sigma_1\sigma_2 c_{12}
\end{aligned}$$

Example: Simplest portfolio

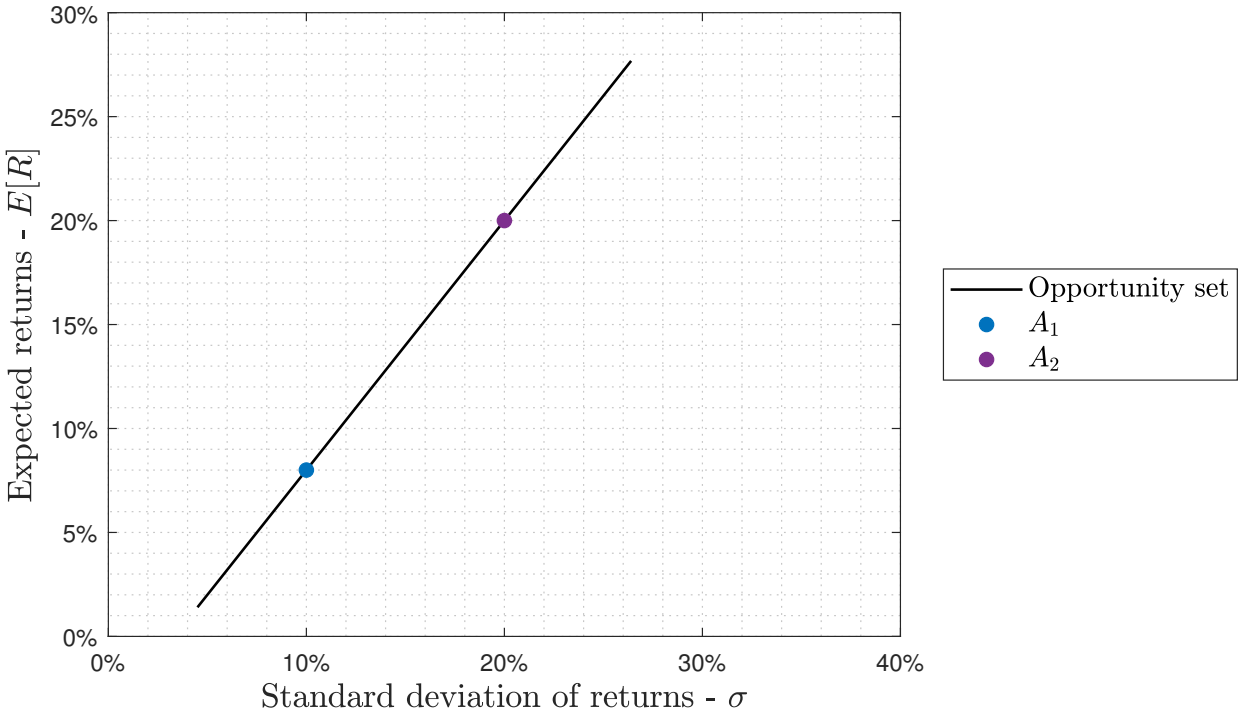
- Suppose that the two assets satisfy the following features:

$$A_1 : \sigma_1 = 10\%, \quad \mathbb{E}[R_1] = 8\%$$
$$A_2 : \sigma_2 = 20\%, \quad \mathbb{E}[R_2] = 20\%$$

- When A_1 and A_2 are perfectly positively correlated, i.e. $c_{12} = 1$, what is the relationship between $\mathbb{E}_p$ and σ_p?
- When A_1 and A_2 are perfectly negatively correlated, i.e. $c_{12} = -1$, what is the relationship between $\mathbb{E}_p$ and σ_p?
- It is typical to plot the portfolios in a $(\mathbb{E}[R], \sigma)$-plane, as is shown below.

Example: Simplest portfolio, cont.

- When A_1 and A_2 are perfectly positively correlated, i.e. $c_{12} = 1$, we have

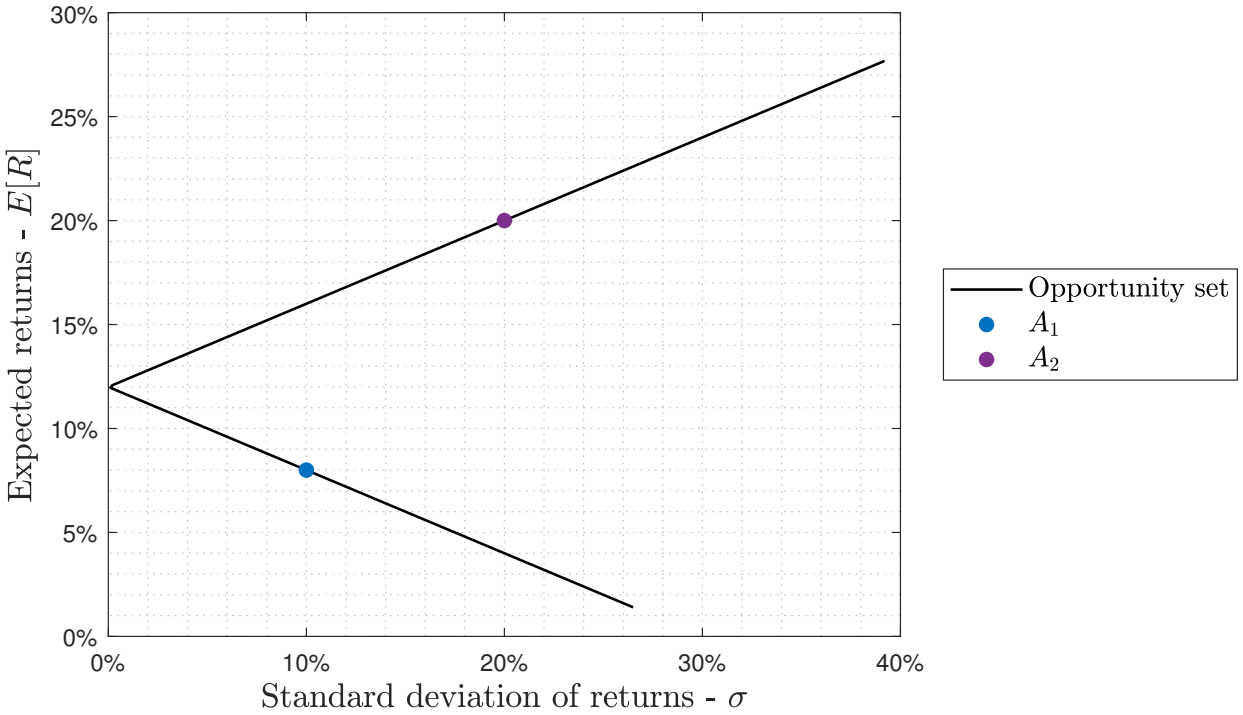


where $\mathbb{E}_p = x\mathbb{E}[R_1] + (1-x)\mathbb{E}[R_2] = 0.2 - 0.12x$ and

$\sigma_p = |x\sigma_1 + (1-x)\sigma_2| = |0.2 - 0.1x|$.

Example: Simplest portfolio, cont.

- When A_1 and A_2 are perfectly negatively correlated, i.e. $c_{12} = -1$, we have

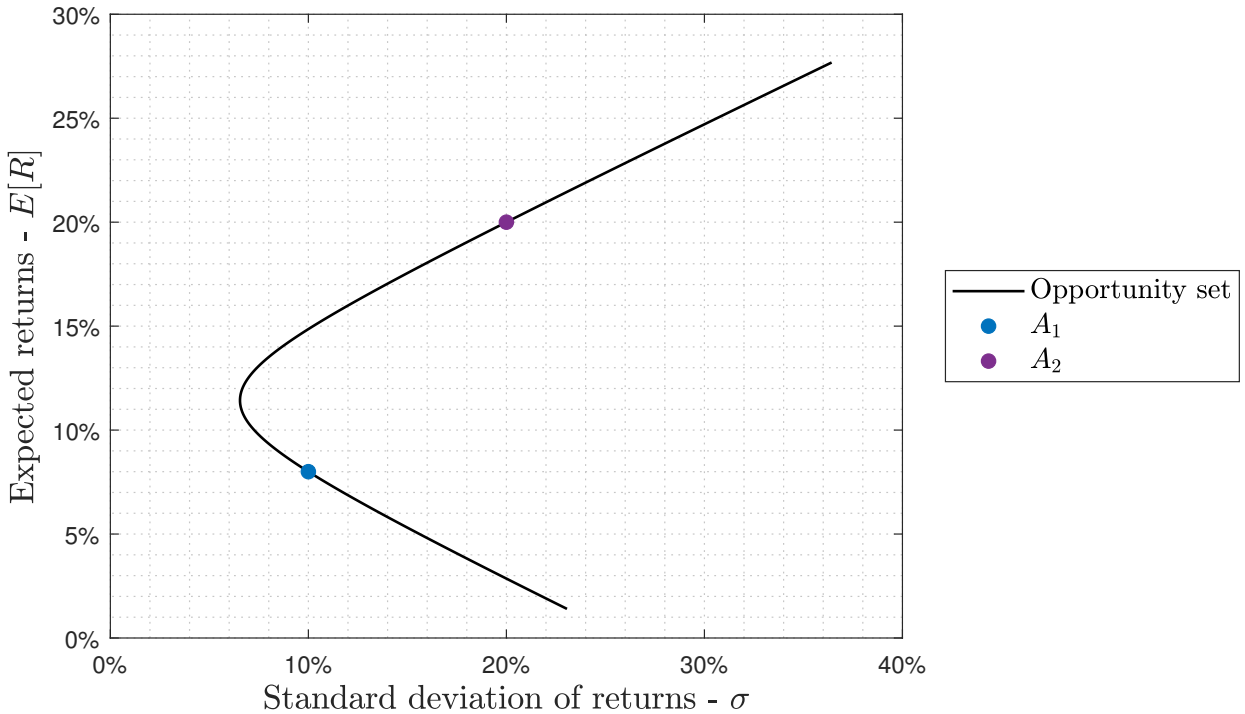


where $\mathbb{E}_p = x\mathbb{E}[R_1] + (1-x)\mathbb{E}[R_2] = 0.2 - 0.12x$ and

$\sigma_p = |x\sigma_1 - (1-x)\sigma_2| = |0.3x - 0.2|$.

Example: Simplest portfolio, cont.

- When the correlation is $c_{12} = -0.5$, we have

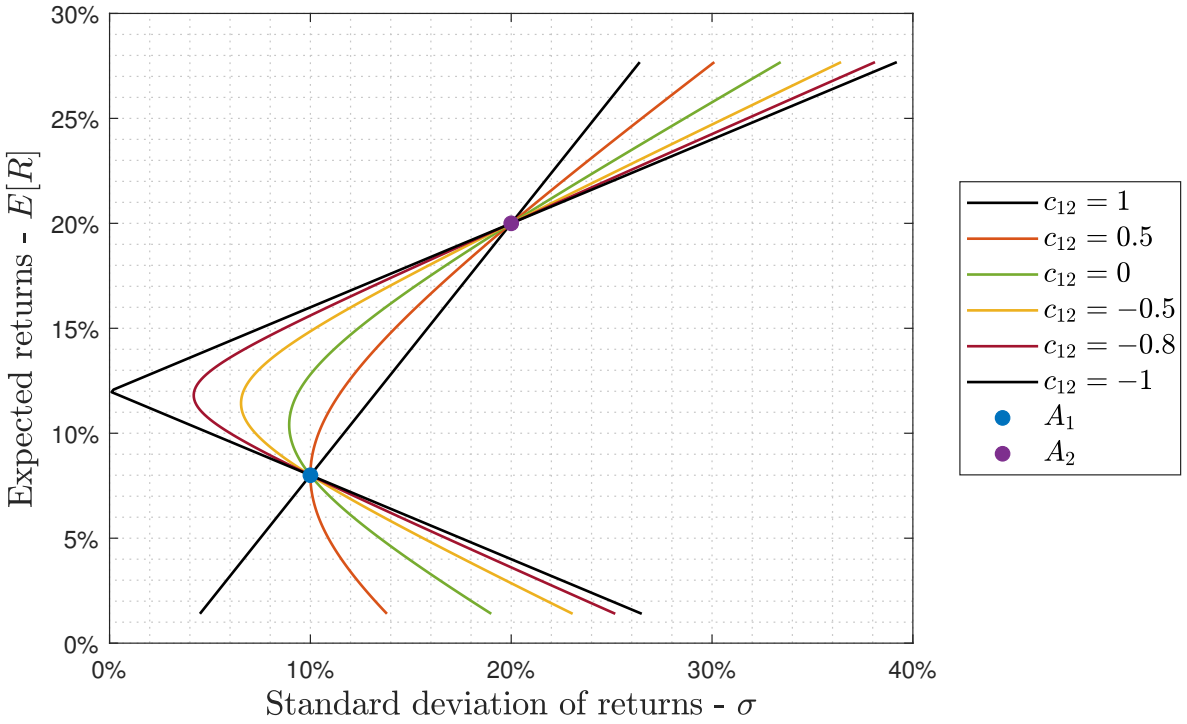


where $\mathbb{E}_p = x\mathbb{E}[R_1] + (1-x)\mathbb{E}[R_2] = 0.2 - 0.12x$ and

$$\begin{aligned}\sigma_p &= \sqrt{0.01x^2 + 0.04(1-x)^2 - 0.02x(1-x)} \\ &= \sqrt{0.07x^2 - 0.1x + 0.04}.\end{aligned}$$

Example: Simplest portfolio, cont.

- When the correlation is $c_{12} \in (-1, 1)$, we have

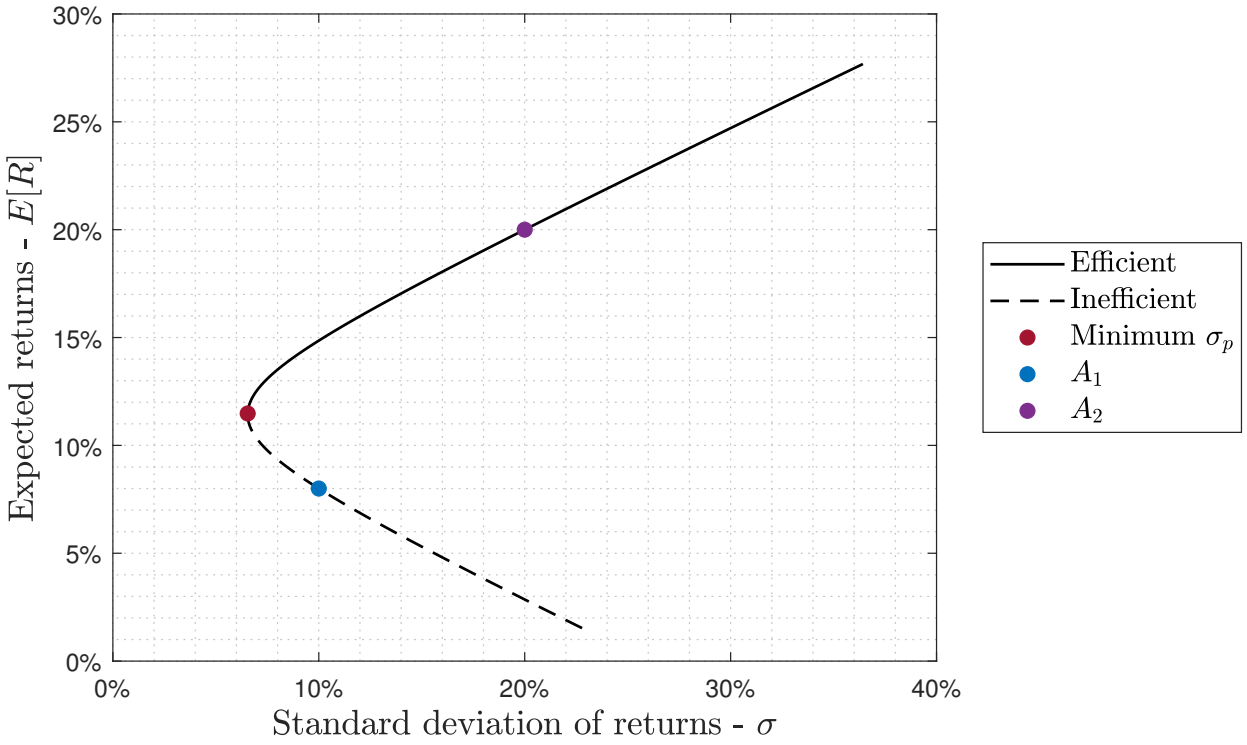


Efficient frontier

- The **efficient frontier** is a portfolio in the opportunity set such that it has the highest possible expected return for a given level of risk, or equivalently has the lowest level of risk for a given level of expected return.
- Investors will only invest in efficient portfolios, given their preferences.

Example: Efficient frontier

- When the correlation is $c_{12} = -0.5$, the solid curve corresponds to a set of efficient frontiers while the dashed curve is a set of inefficient portfolios.



where the minimum variance portfolio is $(x_1, x_2) = \left(\frac{5}{7}, \frac{2}{7}\right)$ with $\sigma_p^2 = \frac{3}{700}$ and $\mathbb{E}[R_p] = \frac{4}{35}$.

Finding the minimum variance portfolio

- In the case of two assets, the weights in the minimum variance portfolio can be found using the following formula,

$$x_1 = \frac{\sigma_2^2 - \sigma_{12}}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}}, \qquad x_2 = 1 - x_1$$

- Then, the minimum variance and expected return are determined by

$$\sigma_p^2 = x_1^2\sigma_1^2 + x_2^2\sigma_2^2 + 2x_1x_2\sigma_{12}$$
$$\mathbb{E}_p = x_1\mathbb{E}[R_1] + x_2\mathbb{E}[R_2]$$

Some important points to consider

- The shape of the opportunity set, and thus the efficient portfolios, depends critically on the correlations between the assets. This is a result of a wider point, which is that the joint behaviour of the assets is now what matters, not just their individual behaviour.
- The lower the value of correlation, the smaller the variance of return of the minimum variance portfolio.
- The curve will also depend on whether short selling is allowed or not: if short selling is not permitted, this restricts the possible set of efficient portfolios.

Question

- How to determine the minimum variance portfolio consisting of n assets $A_1, \cdots, A_n$?

Minimum variance portfolio consisting of n assets

- We aim to find a portfolio within the opportunity set that minimises σ_p^2, while adhering to the budget constraint $\sum_{i=1}^{n} x_i = 1$, where x_i represents the weight invested in asset A_i.
- x_i, $i = 1, \cdots, n$ are the variables we are trying to determine to satisfy our conditions.

$$\begin{aligned} \min_{x_1, \cdots, x_n} \quad & \sigma_p^2 \\ \text{Subject to} \quad & \sum_{i=1}^{n} x_i = 1 \end{aligned}$$

- To solve this constrained optimisation problem, we use the method of Lagrange multipliers.

Minimum variance portfolio consisting of n assets, cont.

- The Lagrangian function is:

$$\mathcal{L}(x_1, \cdots, x_n, \mu) = \sigma_p^2 - \mu \left(\sum_{i=1}^{n} x_i - 1 \right)$$
$$= \sum_{i=1}^{n} x_i^2 \sigma_i^2 + 2 \sum_{i=1}^{n} \sum_{j=i+1}^{n} x_i x_j \sigma_{ij} - \mu \left(\sum_{i=1}^{n} x_i - 1 \right)$$

with partial derivatives

$$\frac{\partial \mathcal{L}}{\partial x_i} = 2 \sum_{j=1}^{n} \sigma_{ij} x_j - \mu, \quad i = 1, \cdots, n$$
$$\frac{\partial \mathcal{L}}{\partial \mu} = - \left(\sum_{i=1}^{n} x_i - 1 \right)$$

Minimum variance portfolio consisting of n assets, cont.

- Setting each of these derivatives equal to zero results in:

$$2\sum_{j=1}^{n}\sigma_{ij}x_j - \mu = 0, \quad i = 1, \cdots, n$$

$$\sum_{i=1}^{n} x_i = 1$$

- These equations can be solved simultaneously to find the required portfolio weights. Linear algebra can help to write these equations in a neat way.

Portfolio consisting of n assets, matrix form

- In the case of n assets, let us define the following vectors,

$$r = \begin{pmatrix} \mathbb{E}[R_1] \\ \vdots \\ \mathbb{E}[R_n] \end{pmatrix}, \quad X = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}, \quad \mathbf{1} = \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}$$

and define the variance-covariance matrix,

$$V = \begin{pmatrix} \sigma_1^2 & \sigma_{12} & \cdots & \sigma_{1n} \\ \sigma_{21} & \sigma_2^2 & \cdots & \sigma_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{n1} & \sigma_{n2} & \cdots & \sigma_n^2 \end{pmatrix}$$

The matrix V is symmetric and positive definite ($X^\top \cdot V \cdot X > 0$) for all $X \neq 0$.

Portfolio consisting of n assets, matrix form, cont.

- Then the expected return and variance of the portfolio are given by
$$\mathbb{E}_p = X^\top \cdot r, \qquad \sigma_p^2 = X^\top \cdot V \cdot X$$

- Both expressions are equivalent to the formulas in page 6.

Finding the minimum variance portfolio, matrix form

- The weights in the minimum variance portfolio can be determined by

$$X_p = \frac{V^{-1} \cdot \mathbf{1}}{\mathbf{1}^\top \cdot V^{-1} \cdot \mathbf{1}}$$

- Then, the expected return and variance of the minimum variance portfolio are calculated by

$$\mathbb{E}_p = X_p^\top \cdot r, \qquad \sigma_p^2 = X_p^\top \cdot V \cdot X_p$$

Example

- Three assets have mean return $\mathbb{E}[R_1] = 0.03$, $\mathbb{E}[R_2] = 0.04$, $\mathbb{E}[R_3] = 0.05$, and the variance-covariance matrix

$$V = \begin{pmatrix} 0.1 & -0.1 & 0 \\ -0.1 & 0.2 & 0 \\ 0 & 0 & 0.1 \end{pmatrix}$$

Assume that short selling is allowed in this market.

i) Find the minimum variance portfolio.

ii) Find the variance of the minimum variance portfolio.

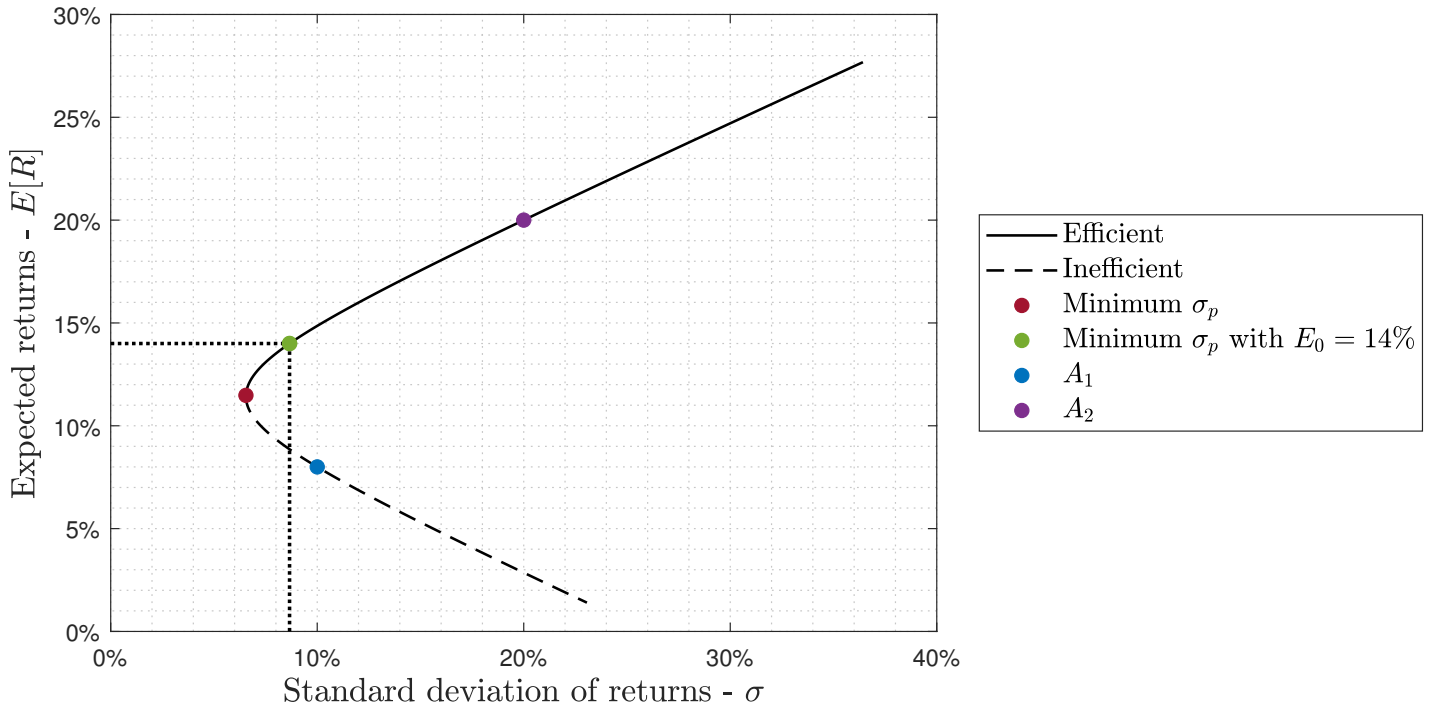
iii) Find the expected return of the minimum variance portfolio.

Question

- For the above examples, no constraints on the expected return were taken into account.
- When investors require the expected return to match a specific target level, how can we calculate the minimum variance portfolio that corresponds to this particular expected return?

Example: Minimum variance portfolio with a required return

- Consider the two assets in page 9 and $c_{12} = -0.5$.



where the minimum variance portfolio with an expected return of 0.14 is $(x_1, x_2) = \left(\frac{1}{2}, \frac{1}{2}\right)$ with $\sigma_p^2 = \frac{3}{400}$.

Minimum variance portfolio with a specific expected return

- We want to find a portfolio within the opportunity set that minimises σ_p^2 for a given expected return such that $\mathbb{E}_p = \mathbb{E}_0$, while adhering to the budget constraint $\sum_{i=1}^{n} x_i = 1$, where x_i represents the weight invested in asset A_i.
- x_i, $i = 1, \cdots, n$ are still the variables we are trying to determine to satisfy our conditions.

$$\begin{aligned} \min_{x_1, \cdots, x_n} \quad & \sigma_p^2 \\ \text{Subject to} \quad & \mathbb{E}_p = \mathbb{E}_0 \\ & \sum_{i=1}^{n} x_i = 1 \end{aligned}$$

Minimum variance portfolio with a specific expected return

- The Lagrangian function is:

$$\mathcal{L}(x_1, \cdots, x_n, \lambda, \mu) = \sigma_p^2 - \lambda(\mathbb{E}_p - \mathbb{E}_0) - \mu\left(\sum_{i=1}^{n} x_i - 1\right)$$
$$= \sum_{i=1}^{n} x_i^2 \sigma_i^2 + 2\sum_{i=1}^{n}\sum_{j=i+1}^{n} x_i x_j \sigma_{ij} - \lambda\left(\sum_{i=1}^{n} x_i \mathbb{E}[R_i] - \mathbb{E}_0\right) - \mu\left(\sum_{i=1}^{n} x_i - 1\right)$$

with partial derivatives

$$\frac{\partial \mathcal{L}}{\partial x_i} = 2\sum_{j=1}^{n} \sigma_{ij} x_j - \lambda \mathbb{E}[R_i] - \mu, \quad i = 1, \cdots, n$$
$$\frac{\partial \mathcal{L}}{\partial \lambda} = -\left(\sum_{i=1}^{n} x_i \mathbb{E}[R_i] - \mathbb{E}_0\right)$$
$$\frac{\partial \mathcal{L}}{\partial \mu} = -\left(\sum_{i=1}^{n} x_i - 1\right)$$

Minimum variance portfolio with a specific expected return

- Setting each of these derivatives equal to zero results in:

$$2\sum_{j=1}^{n}\sigma_{ij}x_j - \lambda\mathbb{E}[R_i] - \mu = 0, \quad i = 1, \cdots, n$$

$$\sum_{i=1}^{n} x_i\mathbb{E}[R_i] = \mathbb{E}_0$$

$$\sum_{i=1}^{n} x_i = 1$$

- Solve these equations to find the required portfolio weights.

Matrix form

- Let $\alpha = \mathbf{1}^\top \cdot V^{-1} \cdot \mathbf{1}$, $\beta = r^\top \cdot V^{-1} \cdot r$ and $\gamma = \mathbf{1}^\top \cdot V^{-1} \cdot r$.
- The weights in the minimum variance portfolio with a specific expected return can be determined by

$$X_p = \frac{1}{\Theta}\left(\alpha \mathbb{E}_0 - \gamma\right)\left(V^{-1} \cdot r\right) - \frac{1}{\Theta}\left(\gamma \mathbb{E}_0 - \beta\right)\left(V^{-1} \cdot \mathbf{1}\right)$$

where $\Theta = \alpha\beta - \gamma^2$.

Example

- Three assets have mean return $\mathbb{E}[R_1] = 0.03$, $\mathbb{E}[R_2] = 0.04$, $\mathbb{E}[R_3] = 0.05$, and the variance-covariance matrix

$$V = \begin{pmatrix} 0.1 & -0.1 & 0 \\ -0.1 & 0.2 & 0 \\ 0 & 0 & 0.1 \end{pmatrix}$$

Assume that short selling is allowed in this market.

i) Find the minimum variance portfolio with an expected return of 0.045.

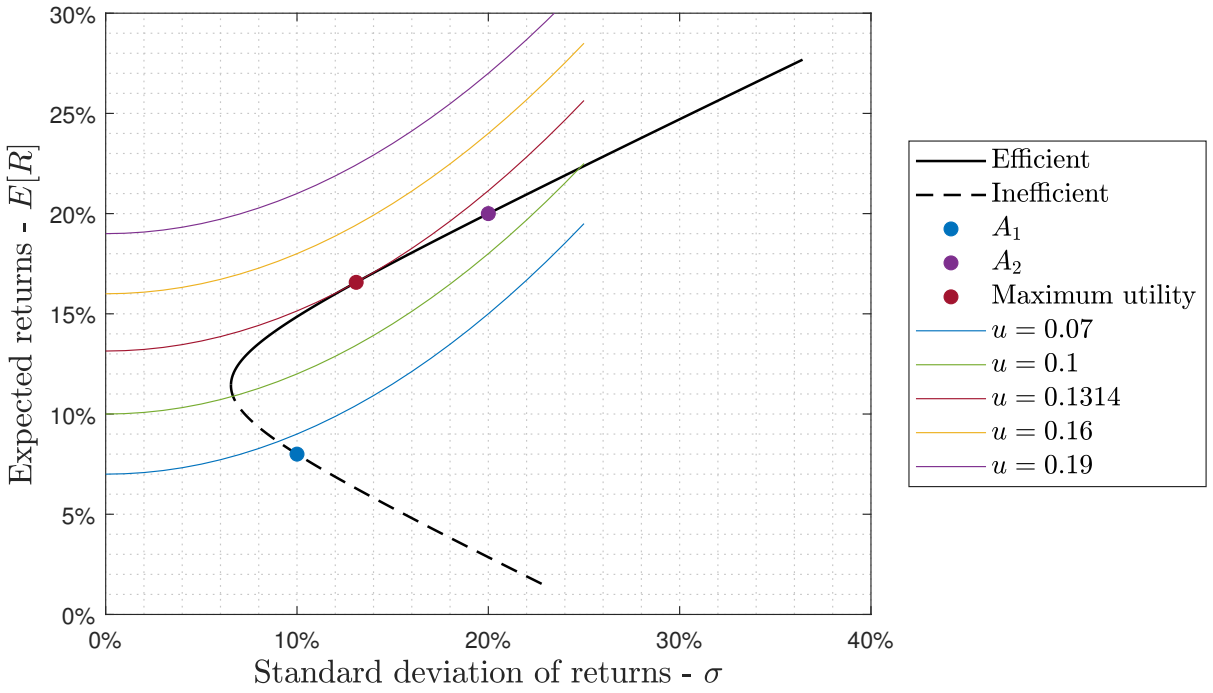
ii) Find the variance of the minimum variance portfolio.

Efficient portfolios with specific preferences

- The analysis that has been presented so far only illustrates how to find efficient portfolios that an investor could choose to invest in. However, the actual selected portfolio will usually depend on the specific preferences of the investor.
- These can also be represented in the $(\mathbb{E}[R], \sigma)$-plane using indifference curves. All points on one indifference curve have the same expected utility value to the investor.
- Note that the indifference curves are upward sloping due to the risk aversion of the investor. In fact, investors want more return as the risk increases, hence upward sloping. The gradient of the upward slope will increase in steepness the more risk averse an investor is.

Efficient portfolios with specific preferences, cont.

- Consider the two assets in page 9 and $c_{12} = -0.5$.
- The efficient portfolio selected will be the one tangent to the highest indifference curve (which will maximise expected utility). In the example below, this is the red curve,



where the utility function is $u = \mathbb{E}_p - 2\sigma_p^2$, and the maximum utility portfolio is $(x_1, x_2) = \left(\frac{2}{7}, \frac{5}{7}\right)$ corresponding to the maximum utility $u = \frac{23}{175} \approx 0.1314$.

Utility function

- The utility function is usually expressed as

$$u = \mathbb{E}_p - \kappa\sigma_p^2$$

where κ is a positive constant.

- Combining the expressions of $\mathbb{E}_p$ and σ_p^2, it can be rewritten as

$$u = \sum_{i=1}^{n} x_i \mathbb{E}[R_i] - \kappa \left(\sum_{i=1}^{n} x_i^2 \sigma_i^2 + 2 \sum_{i=1}^{n} \sum_{j=i+1}^{n} x_i x_j \sigma_{ij} \right)$$

where $\sum_{i=1}^{n} x_i = 1$.

Maximizing the utility

- Therefore, we want to find a portfolio within the opportunity set that maximises u, while adhering to the budget constraint $\sum_{i=1}^{n} x_i = 1$, where x_i represents the weight invested in asset A_i.
- x_i, $i = 1, \cdots, n$ are still the variables we are trying to determine to satisfy our conditions.

$$\begin{aligned} \max_{x_1, \cdots, x_n} \quad & \mathbb{E}_p - \kappa \sigma_p^2 \\ \text{Subject to} \quad & \sum_{i=1}^{n} x_i = 1 \end{aligned}$$

Maximizing the utility, cont.

- To maximize the utility, the Lagrangian function is:

$$\mathcal{L}(x_1, \cdots, x_n, \mu) = \mathbb{E}_p - \kappa\sigma_p^2 - \mu\left(\sum_{i=1}^{n} x_i - 1\right)$$
$$= \sum_{i=1}^{n} x_i \mathbb{E}[R_i] - \kappa\left(\sum_{i=1}^{n} x_i^2\sigma_i^2 + 2\sum_{i=1}^{n}\sum_{j=i+1}^{n} x_i x_j \sigma_{ij}\right) - \mu\left(\sum_{i=1}^{n} x_i - 1\right)$$

with partial derivatives

$$\frac{\partial \mathcal{L}}{\partial x_i} = \mathbb{E}[R_i] - 2\kappa\sum_{j=1}^{n}\sigma_{ij}x_j - \mu, \quad i = 1, \cdots, n$$
$$\frac{\partial \mathcal{L}}{\partial \mu} = -\left(\sum_{i=1}^{n} x_i - 1\right)$$

Maximizing the utility, cont.

- Setting each of these derivatives equal to zero results in:

$$2\kappa \sum_{j=1}^{n} \sigma_{ij} x_j + \mu = \mathbb{E}[R_i], \quad i = 1, \cdots, n$$

$$\sum_{i=1}^{n} x_i = 1$$

- Solve these linear equations to find the weights of the maximum utility portfolio.

Maximizing the utility, matrix form

- The weights in the maximum utility portfolio can be determined by

$$X_p = \frac{V^{-1} \cdot r}{2\kappa} - \frac{(\gamma - 2\kappa)\left(V^{-1} \cdot \mathbf{1}\right)}{2\kappa\alpha}$$

where $\alpha = \mathbf{1}^\top \cdot V^{-1} \cdot \mathbf{1}$ and $\gamma = \mathbf{1}^\top \cdot V^{-1} \cdot r$.

Example

- Three assets have mean return $\mathbb{E}[R_1] = 0.03$, $\mathbb{E}[R_2] = 0.04$, $\mathbb{E}[R_3] = 0.05$, and the variance-covariance matrix

$$V = \begin{pmatrix} 0.1 & -0.1 & 0 \\ -0.1 & 0.2 & 0 \\ 0 & 0 & 0.1 \end{pmatrix}$$

  Given the utility function $u = \mathbb{E}_p - \frac{1}{3}\sigma_p^2$ and assuming that short selling is allowed in this market, find the maximum utility portfolio.